

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

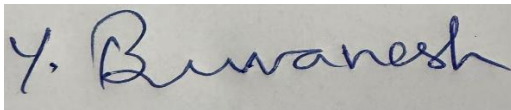
Total Marks: 25

Code of Conduct

I Buvanesh Y (192372367) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. Video Streaming Platform

Architecture Overview:

Videos are uploaded to object storage (Amazon S3 or Google Cloud Storage). A transcoding pipeline (AWS Elemental MediaConvert) converts each video into multiple resolutions — 360p, 720p, 1080p, 4K — using adaptive bitrate formats like HLS or MPEG-DASH. The player automatically switches quality based on the user's internet speed, preventing buffering. All encoded segments are delivered through a CDN (CloudFront, Akamai, or Fastly) to users worldwide.

Caching Strategy:

CDN edge servers cache popular video chunks at locations near users — someone in India gets content from a Mumbai edge node, not a US origin server. This serves 80-90% of all traffic without touching origin. An Origin Shield sits between the CDN and S3, collapsing multiple cache-miss requests into one. Popular content (trending shows, live events) is pre-warmed on CDN nodes before going live, so the first user doesn't experience a slow load.

Load Balancing & Auto-Scaling:

An Application Load Balancer (ALB) distributes requests across streaming API servers using round-robin or least-connections algorithms. It runs health checks every 30 seconds — unhealthy instances are automatically removed from the pool. Auto Scaling Groups monitor CPU utilization and active connections. When a major live event causes a 10x traffic spike, new EC2 instances or containers launch within 2-3 minutes. When traffic drops, unused instances are terminated to avoid waste.

Cost Optimization:

- Reserved instances for predictable baseline load (40-60% savings over on-demand)
- Spot instances for transcoding jobs — 70-90% cheaper, and failed jobs simply retry
- S3 Intelligent Tiering automatically moves unwatched old content to cheaper storage classes
- Compression (Brotli/gzip) on manifests reduces CDN data transfer costs
- CDN volume discounts negotiated for high-traffic platforms

2. Multi-Cloud Container Orchestration (Fintech)

Container Platform:

Every application is packaged as a Docker container, ensuring identical behavior across development, staging, and production. Kubernetes manages these containers — EKS on AWS, AKS on Azure, GKE on GCP. A unified control plane like Rancher or Google Anthos provides a single dashboard to deploy and manage workloads across all clouds without switching tools.

Consistent Deployments with GitOps:

Helm charts define each application as a versioned, reusable package that deploys identically on any cluster. ArgoCD or Flux continuously watch your Git repository — the moment you push a new version,

it automatically syncs to all clusters. Every environment always matches what's in Git. Rollbacks are instant — just revert the commit.

Service Discovery & Scaling:

Kubernetes internal DNS lets services find each other by name without hardcoded IPs. A service mesh (Istio) extends this across clouds with automatic mTLS encryption between all services — critical for fintech compliance. It also handles traffic routing, circuit breaking (stops cascading failures), and provides full request tracing for debugging. Horizontal Pod Autoscaler scales pods based on CPU/memory. KEDA scales based on business events like payment queue depth.

Benefits vs Risks:

Benefits	Risks
No vendor lock-in	Much higher operational complexity
Pick best service from each cloud	Data egress costs between clouds
Survive a complete cloud provider outage	Inconsistent security policies across clouds
Better pricing negotiation leverage	Requires skilled multi-cloud engineers
Meet data residency compliance per region	Harder debugging across environments

3. Backup & Disaster Recovery (E-Learning)

Versioning & Deletion Protection:

Enable versioning on every S3 bucket storing course videos, assignments, and user uploads. When a file is deleted, a delete marker is created — the original file is preserved and can be restored in seconds. Object Lock (WORM mode) prevents permanent deletion during a retention period, protecting against both accidental deletion and ransomware attacks. Even administrators cannot bypass this during the lock period.

Database Backup Strategy:

Continuous WAL (Write-Ahead Log) archiving streams transaction logs to S3 every 5 minutes, enabling point-in-time recovery to any exact moment. Automated daily snapshots are retained for 30 days, weekly snapshots for 1 year. All snapshots are automatically copied to a second region — a full regional disaster cannot destroy both copies simultaneously.

Cross-Region Replication:

S3 Cross-Region Replication continuously syncs every object to a bucket in another region within minutes. RDS cross-region read replicas keep the database synchronized. A warm standby environment — a scaled-down production copy — runs 24/7 in the backup region, ready to scale up and accept traffic at any time.

RTO & RPO Targets:

- **RPO (max data loss):** Continuous replication and 5-minute WAL archiving limits data loss to under 15 minutes in the worst case.
- **RTO (recovery speed):** With a warm standby, you're back online in 15-30 minutes. Route 53 failover policy automatically redirects users when health checks detect primary region failure.

Business Continuity Steps:

1. Route 53 detects primary failure → switches DNS to standby region automatically
2. Standby database promoted to primary, accepting write traffic
3. Auto Scaling brings standby application servers to full production capacity
4. Students experience a brief interruption but all course progress and grades are fully intact
5. Quarterly DR drills are mandatory — a backup that has never been tested cannot be trusted

4. Hybrid Cloud for Government

Architecture Design:

Sensitive data (citizen PII, classified records, regulated databases) stays strictly on-premises under physical security controls. Non-sensitive workloads (public-facing websites, analytics, document management, email) move to a government-certified cloud like AWS GovCloud or Azure Government — these are FedRAMP certified and physically isolated from commercial regions. The two environments connect via AWS Direct Connect or Azure ExpressRoute — dedicated private fiber links with guaranteed bandwidth and predictable latency, never traversing the public internet. A backup VPN tunnel provides redundancy if the primary link fails.

Identity & Access Management:

On-premises Active Directory federates with cloud IAM (Azure AD or AWS IAM Identity Center) using SAML/OIDC. Employees use a single set of credentials for both environments — no separate cloud passwords. MFA is enforced everywhere without exception. Role-Based Access Control is defined centrally — a finance analyst gets access only to finance systems, whether on-premises or cloud. Privileged Identity Management enforces just-in-time admin access — temporary, time-limited, approval-required, and fully audited.

Encryption Strategy:

AES-256 encryption for all data at rest on both sides. TLS 1.3 for all data in transit, including the Direct Connect link (encrypted with MACsec at the network layer). Encryption keys are managed by an on-premises Hardware Security Module (HSM) — the cloud provider never holds master keys. Cloud services use BYOK (Bring Your Own Key), giving the agency complete cryptographic control over all cloud-stored data.

Key Challenges:

- Applications split across environments experience higher latency — tightly coupled services must stay in the same environment
- Enforcing consistent firewall rules and patch policies across two platforms requires tools like Azure Arc or AWS Systems Manager
- Unified monitoring across on-premises and cloud requires a central SIEM (Splunk or Microsoft Sentinel)
- Staff need expertise in both traditional data center operations and cloud platforms — a significant training investment
- Cloud cost forecasting is harder when combined with fixed on-premises infrastructure costs

5. Multi-Region High Availability (SaaS)

Deployment Strategy:

Run Active-Active across two or more regions. Both regions serve live production traffic simultaneously. A global load balancer (AWS Route 53 with latency-based routing, or Cloudflare) sends each user to their nearest healthy region for lowest latency. If one region fails completely, the global load balancer detects it within seconds and shifts 100% of traffic to the healthy region — no manual intervention, no failover script to run, minimal user impact.

Data Replication:

Amazon Aurora Global Database or CockroachDB replicates data cross-region with sub-second lag. Each region reads from its local database copy (low latency reads). Writes go to the primary region and replicate everywhere within a second. On primary region failure, a replica promotes to primary in under 60 seconds with near-zero data loss. All application servers must be completely stateless — user sessions stored in Redis (replicated cross-region) so any server in any region handles any request seamlessly.

Failover Configuration:

Route 53 health checks run every 10 seconds testing real application functionality — not just a ping, but actually verifying login and data retrieval work. Three consecutive failures trigger automatic DNS failover. Run chaos engineering exercises quarterly using AWS Fault Injection Simulator — deliberately kill an entire region in staging and verify that failover behaves exactly as designed. If you've never tested it, you don't know if it works.

Monitoring & Alerting:

Track the four golden signals for each region independently: Latency (p95, p99), Error Rate, Traffic Volume, and Saturation. Synthetic monitoring runs automated user journey simulations every minute from multiple global locations — it catches a broken checkout flow before any real customer notices. SLO dashboards track error budget consumption in real time — teams escalate when the budget burns too fast, not after it's fully gone.

Alert tiers ensure the right urgency:

- **P1:** Region health check failing, error rate above 5%, replication lag over 30 seconds → immediate PagerDuty page to on-call engineer
- **P2:** Latency degraded, single AZ unhealthy, disk above 80% → 30-minute response window
- **P3:** Non-critical trends, cost anomalies → next business day

Every alert links to a runbook with exact resolution steps, so on-call engineers can respond confidently at 3am without needing to escalate immediately.